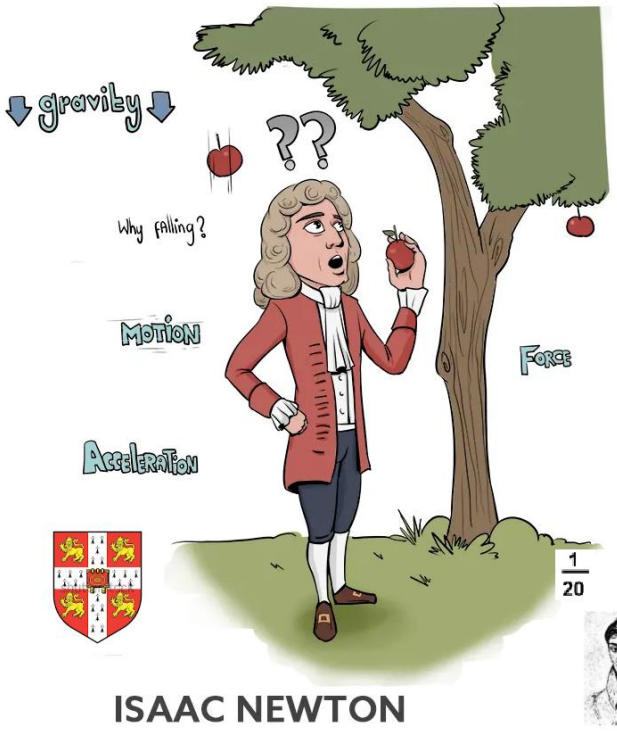
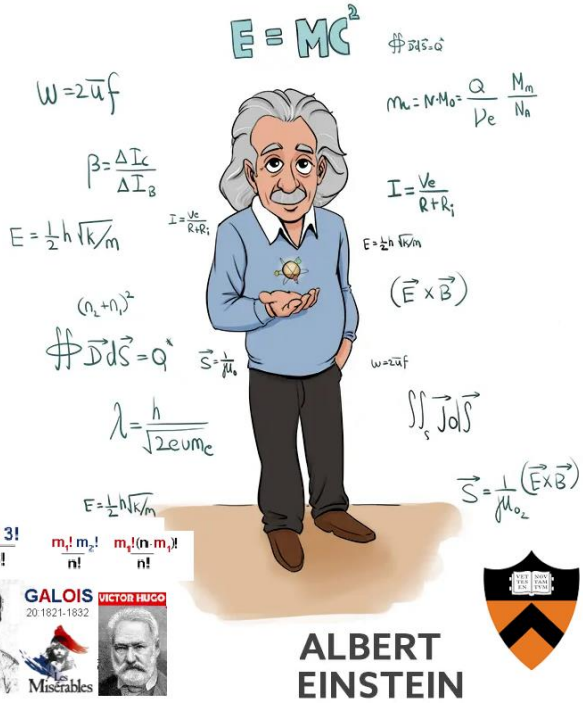




School gravity MOTION ==formalism==> University $E = MC^2$ $\oint \vec{D}\cdot d\vec{S}$ $\iint_S \vec{J}\cdot d\vec{S}$

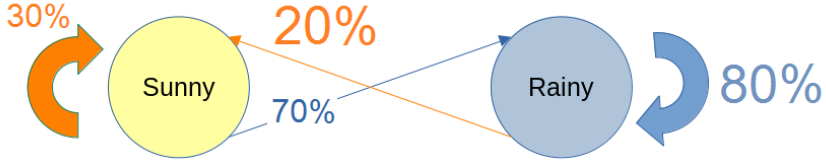
CONCRETE AND ABSTRACT THINKING







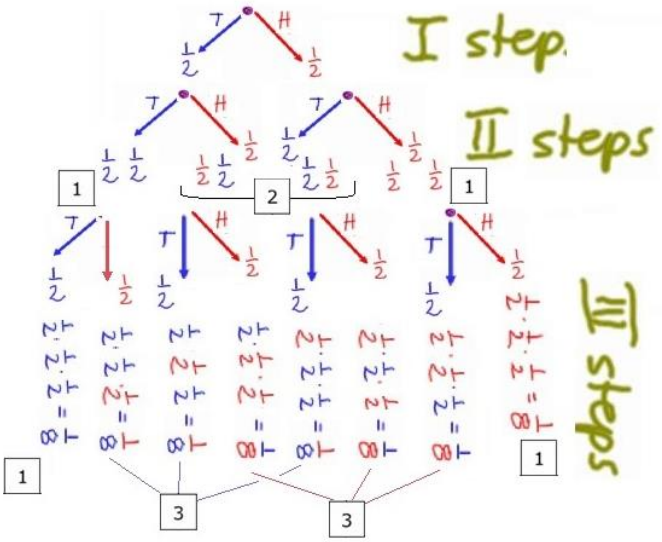
Markoff Chain Probability Model for Oxford Weather





Pascal's triangle

1								← 1
1	1							← 2
1	2	1						← 4
1	3	3	1					← 8
1	4	6	4	1				← 16
1	5	10	10	5	1			← 32
1	6	15	20	15	6	1		← 64
1	7	21	35	35	21	7	1	← 128



$$(a+b)^0 = 1$$
$$(a+b)^1 = a+b$$
$$(a+b)^2 = a^2+2ab+b^2$$
$$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$$
$$(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+b^4$$
$$(a+b)^5 = a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$$

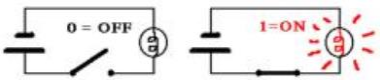


Massachusetts Institute of Technology (MIT)



Lecture by Pr. Bob Gallagher
Boole (1815-1864) & **Shannon** (1916-2001)



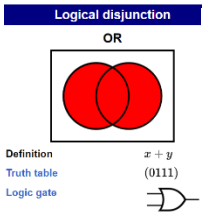




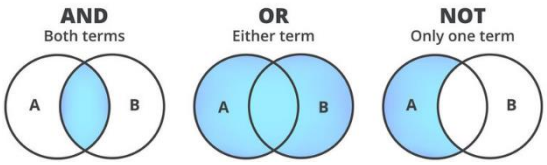
Logical addition (disjunction)

A	B	F=A∨B
0	0	0
0	1	1
1	0	1
1	1	1

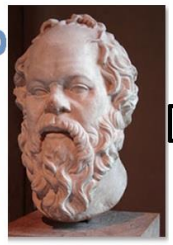
A	B	$A \vee B$
True	True	True
True	False	True
False	True	True
False	False	True



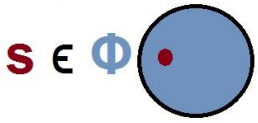
BOOLEAN LOGIC

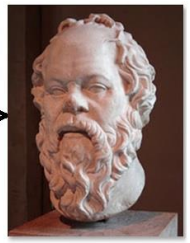


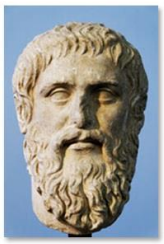
Good logic

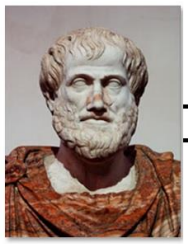

Socrates

Socrates was a philosopher

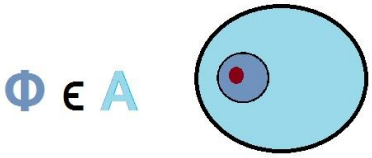



Socrates


Plato

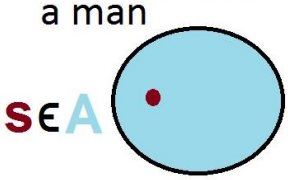

Aristotle

philosophers are men



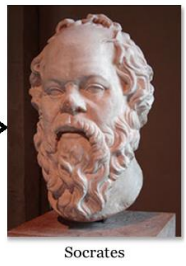


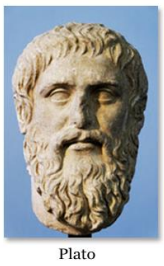
Socrates was a man

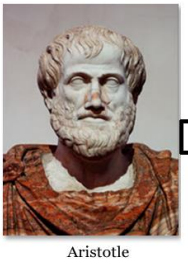


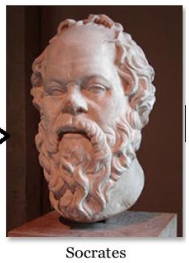
Bad logic



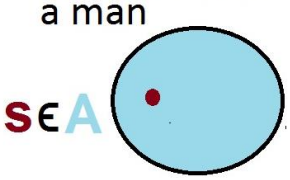

Socrates


Plato

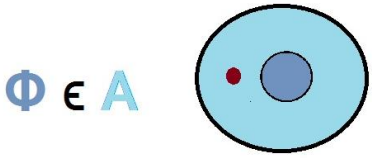

Aristotle


Socrates

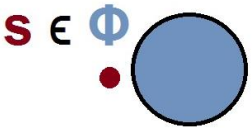
Socrates was a man

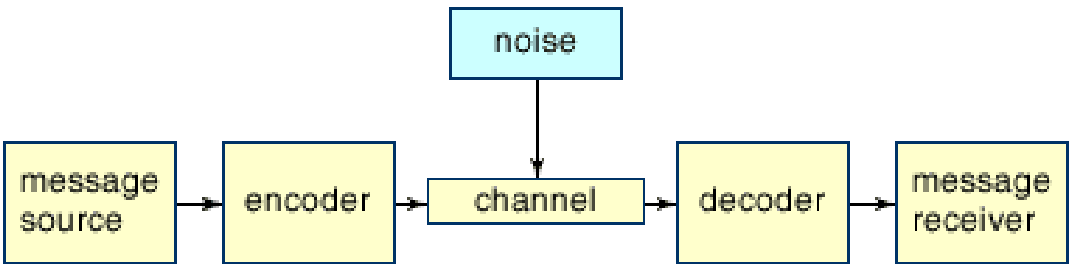


philosophers are men



Socrates was a philosopher

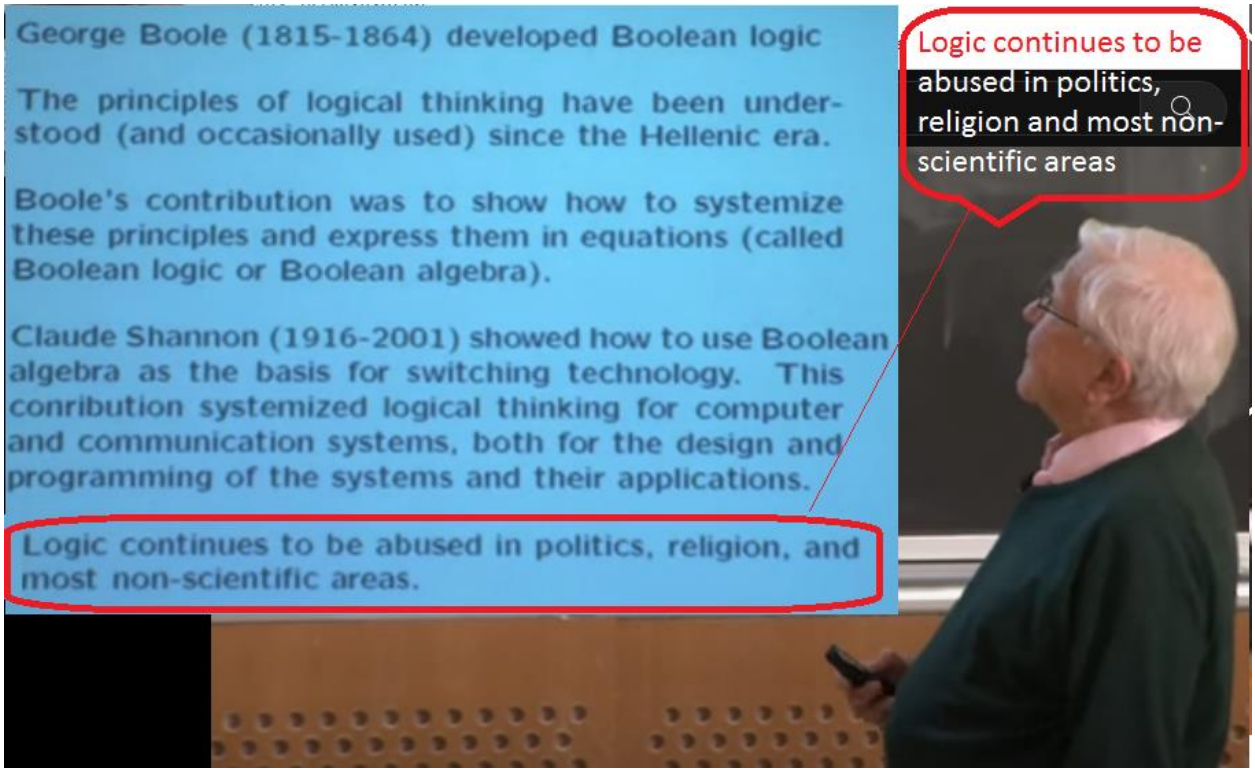


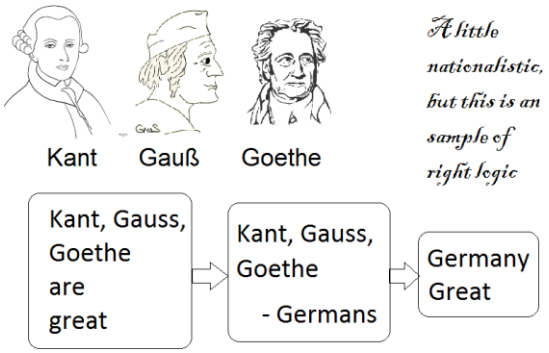


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Resume of Lecture by Pr. Bob Gallagher from MIT

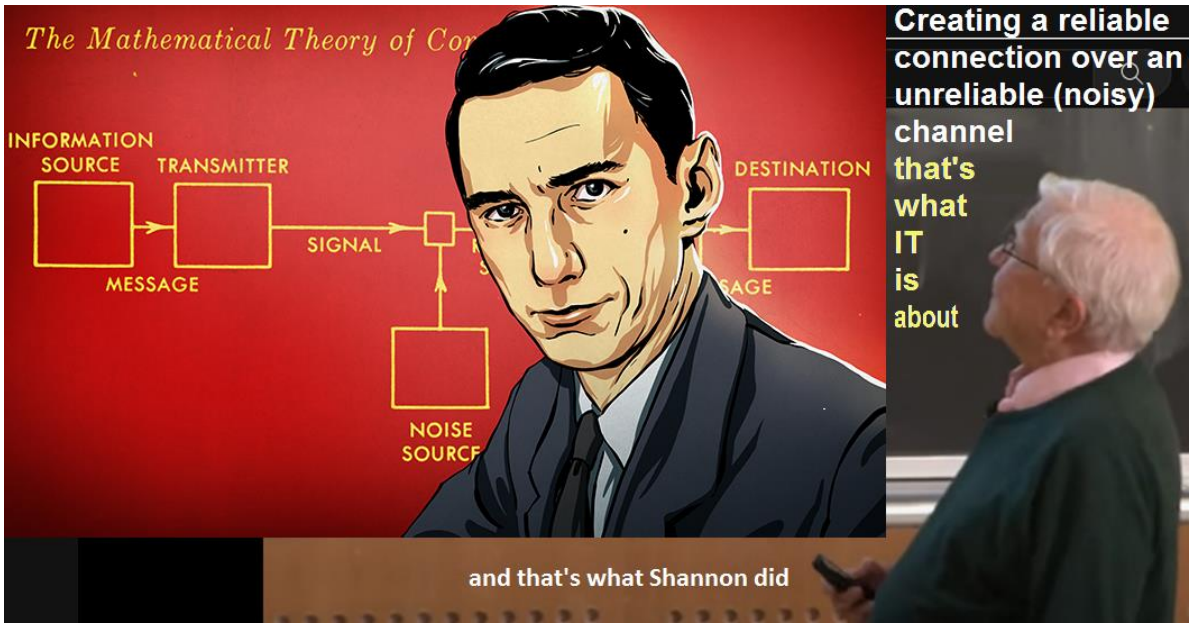
Massachusetts Institute of Technology (MIT)

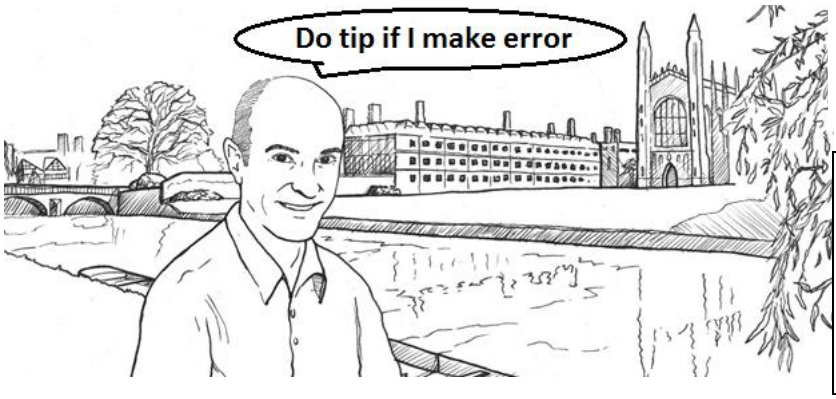




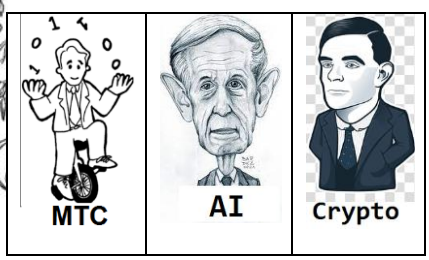


Bad logic (abuse of logic)





Sir **Dr. D. MacKay**,
University of Cambridge
(22 April 1967 – 14 April 2016)

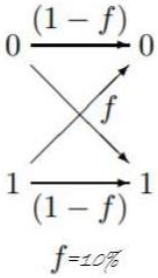


"I believe in clean energy,
but I also believe in mathematics"

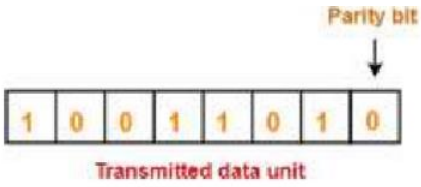


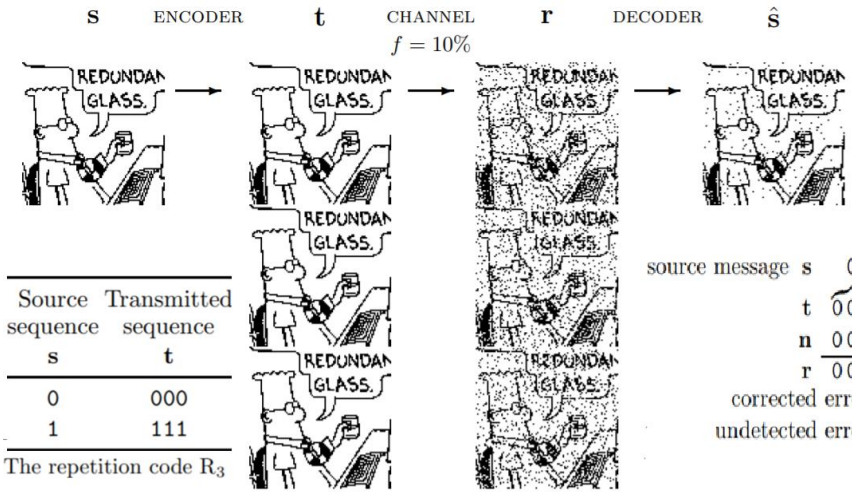












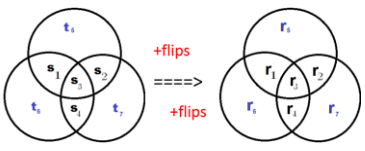
Source sequence s	Transmitted sequence t
0	000
1	111

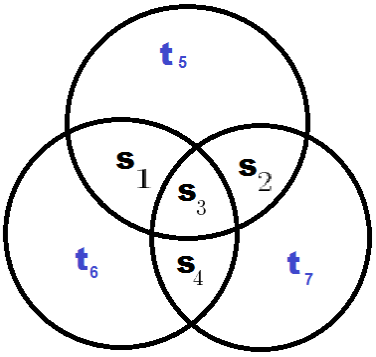
The repetition code R_3

source message s	0	0	1	0	1	1	0
t	000	000	111	000	111	111	000
n	000	001	000	000	101	000	000
r	000	001	111	000	010	111	000
corrected errors		⋆					
undetected errors					⋆		

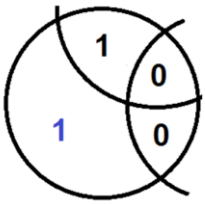
7.4. Hamming code.

$$\frac{4}{\Sigma} \rightarrow \frac{7}{t}$$



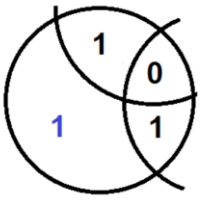


satisfied

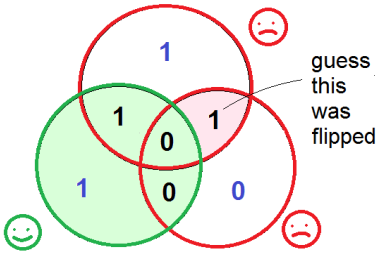


1+1+0+0=**0**

not satisfied



1+1+0+1=**1**





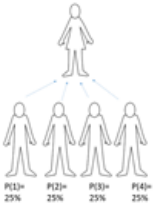
Say **NO** to the **first**



Say **YES** to the **second** if it is better than the first



Say **NO** to the **third** only if it is worse than all the others



1	2	3	4

1 question - the first or second group

1	2	3	4
X			Question 1

2 question - the first or second group

1	2		
	X		Question 2

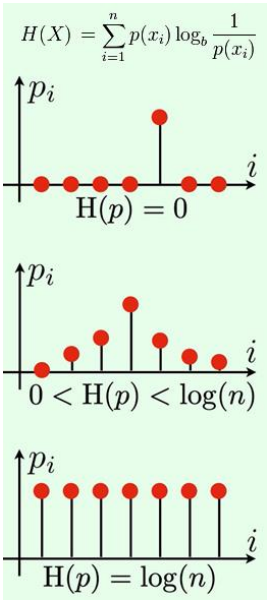
Average number of questions =

2*0.25+ **2***0.25+ **2***0,25+ **2***0,25 =**2**

Average number of questions =

1*0.5+	2*0.25+	3*0,125+	3*0,125
			

Question 1. Is this Zuckerberg?	 50%	1*0.5
Question 2. Is this Sergey Brin?	 25%	2*0.25
Question 3. Is this Stefan from BMW?	 12,5%	3*0,125
So Prince Saud	 12,5%	3*0,125
Average number of questions =		1,75



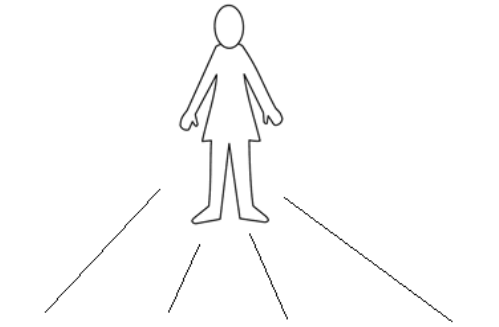
$$\sum_{i=1}^{n} p(i) \log_2 \frac{1}{p(i)}$$

Quantifying information

$$I(x_i) = \log_2\left(\frac{1}{p_i}\right)$$

number of bits required to encode choice

$$\sum_{i=1}^{n} p(x_i)\, I(x_i)$$





Mark Zukerberg P(1)= **50%**



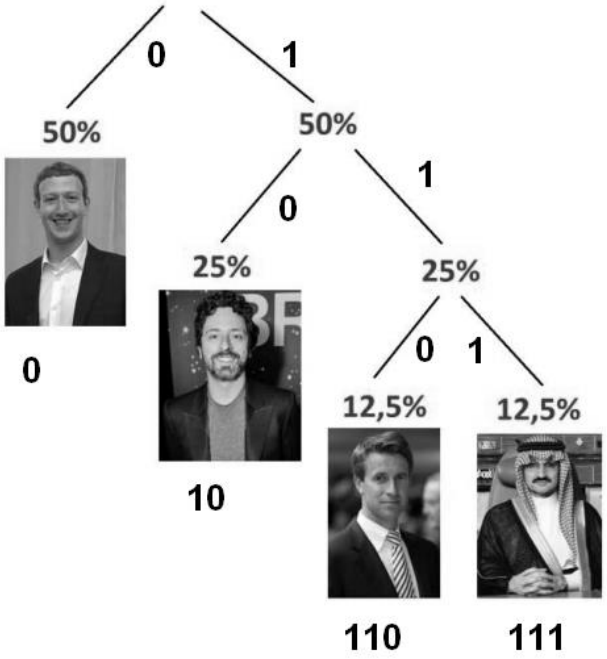
Sergey Brin P(2)= **25%**



Stefan Quandt P(3)= **12,5%**



Prince Al Saud P(4)= **12,5%**



First-order approximation (symbols independent but with frequencies of Belarusian txt).

Мама мыла ра

М - 3	30%	1-3	М
а - 4	40%	4-7	а
ы -1	10%	8	-ы
л - 1	10%	9	-л
р - 1	10%	10	-р

10

лл а мам ма р



Мама мыла ра

Ма - 2	22%	1-2	ма
ам - 2	22%	3-4	ам
мы - 1	11%	5	мы
ыл - 1	11%	6	ыл
ла - 1	11%	7	ла
а р - 1	11%	8	ар
ра - 1	11%	9	ра

9

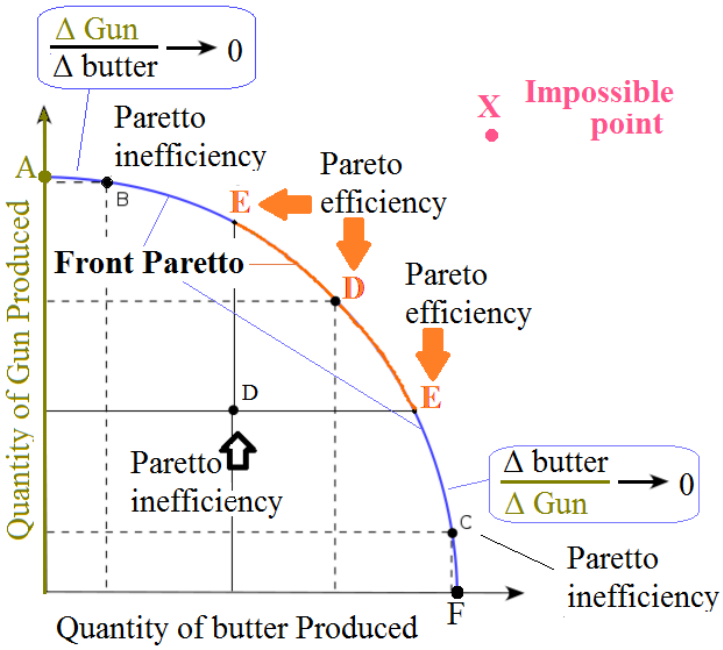
0. 4 6 7 3 1 9 1 6 7 3 5

ам ыл ла ам ма ра ма ыл ла ам мы

мылла рама

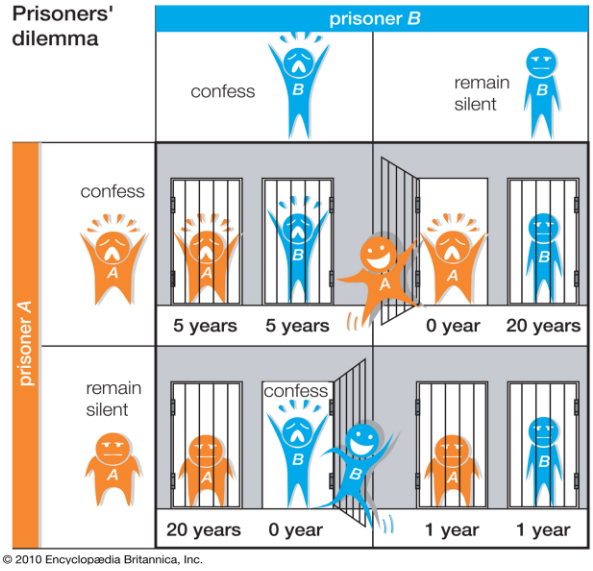


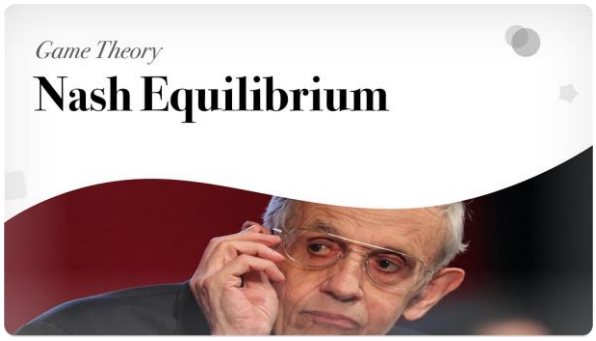
Second-order approximation (digram (2-symbols) structure as in Belarusian)





The orange sector E-D-E is the most Pareto efficient - since an increase in one indicator leads to a decrease in another.

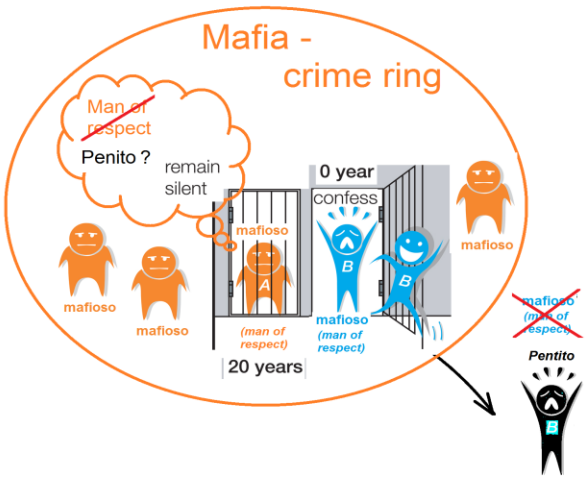




** => Nash equilibrium

$H_2(x)$ / $H_1(x)$	Player 2	Recognition;	Non-recognition;
Player 1		1	2
Recognition;	1	* -5 / -5 *	-20 / 0
Non-recognition;	2	0 / -20	-1 / -1

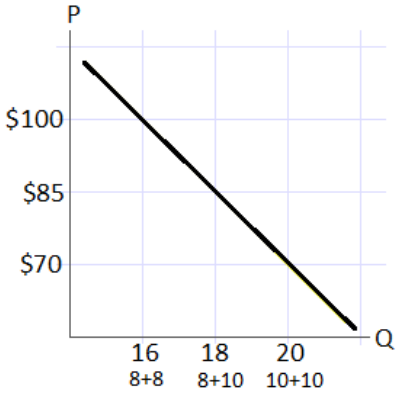
-1-1 Pareto Optimality



Oil price hits 18-year low

Brent crude, US dollars per barrel





Demand curve

	Barrel	1.	2.
		$8 \cdot 10^6$ barrels/day	$10 \cdot 10^6$ barrels/day
1.	$8 \cdot 10^6$ barrels	$800 millions per day $100 per barrel $800 millions per day	$850 millions per day $85 per barrel $680
2.	$10 \cdot 10^6$ barrels/day	$680 millions per day $85 per barrel $850 millions per day	$700 millions per day $70 per barrel $700 millions per day

